
Jev
Course

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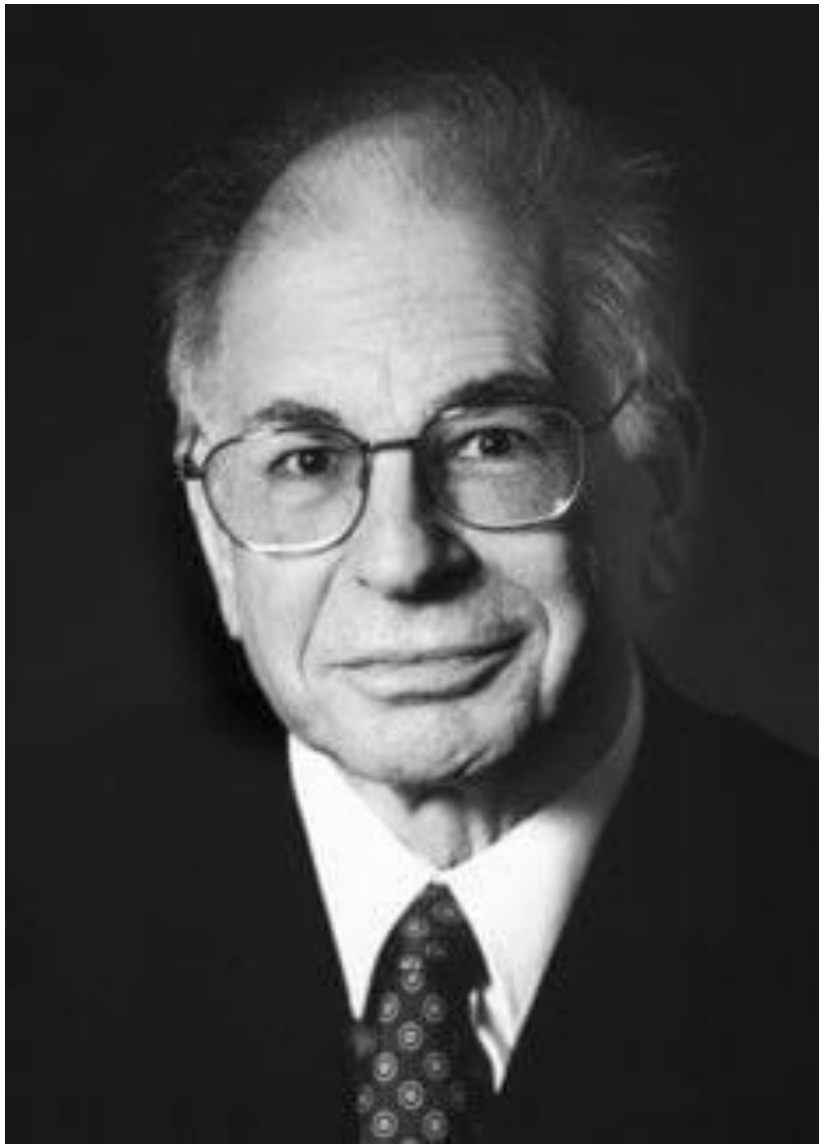
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# Daniel Kahneman

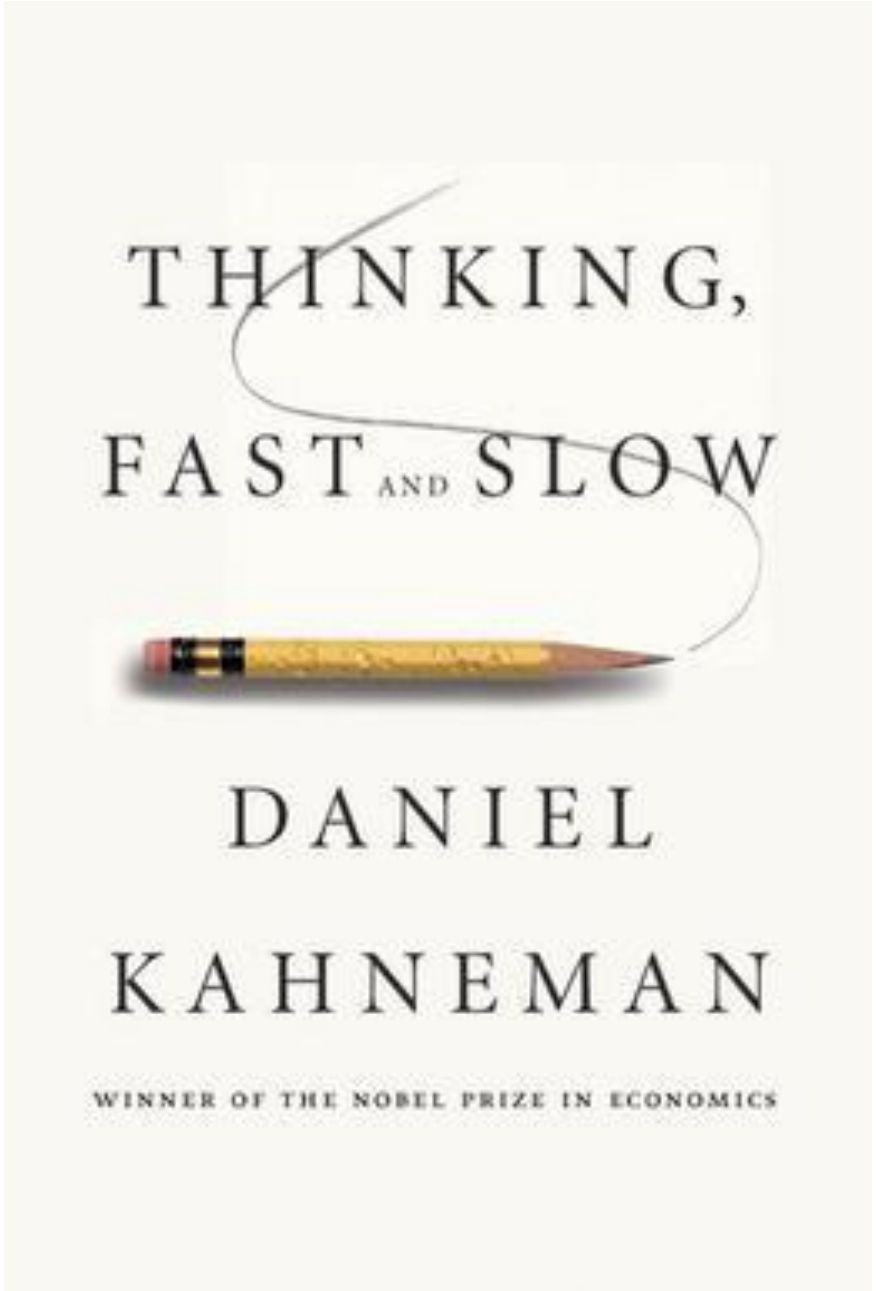
Daniel Kahneman was an Israeli-American psychologist best known for his work on the psychology of judgment and decision-making as well as behavioral economics, for which he was awarded the 2002 Nobel Memorial Prize in Economic Sciences together with Vernon L. Smith.



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**2 + 2 =**

**17 * 24 =**



# Thinking, Fast and Slow × Jev AI

Two Systems of Thinking and How Jev Brings Structured, Probabilistic Decisions

## System 1

**Fast, Automatic, Intuitive**



- Fast
- Automatic
- Emotional
- Often unconscious
- Uses mental shortcuts (heuristics)

## Jev AI

**System One for Structured Decisions**



| | |
|---|---|
| Option A | 0.72 |
| Option B | 0.18 |
| Option C | 0.10 |

- Typed output (labels)
- Probabilistic confidence
- Consistent and testable
- Easy to integrate into workflows

## System 2

**Slow, Deliberate, Logical**



- Slow
- Effortful
- Logical and analytical
- Conscious
- Used for complex and novel problems

# The Hidden Danger of Intuition

Cognitive biases can distort human judgment — Jev AI provides structured, testable decisions.

## Human Intuition

Fast judgment, but prone to bias




### Anchoring Bias

The initial price influences the judgment, even if a better price is available.



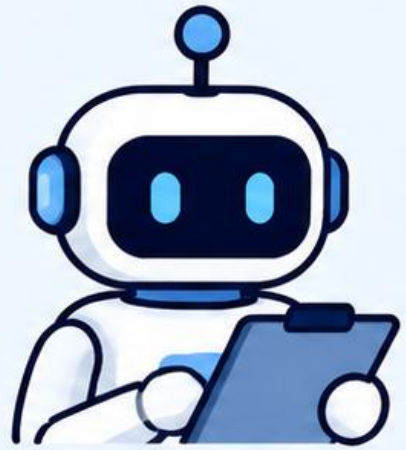

### Availability Bias

Recent news makes rare events feel more likely than they actually are.

**May lead to a distorted decision**

## Jev AI

Structured evaluation with predefined criteria



### Evaluate with clear criteria

| | |
|---|---|
| Market value | 40% |
| Location | 25% |
| Property condition | 20% |
| Neighborhood | 15% |

### Probabilistic decision

| | |
|---|---|
| Buy | 0.72 |
| Wait | 0.20 |
| Avoid | 0.08 |

### Testable and transparent

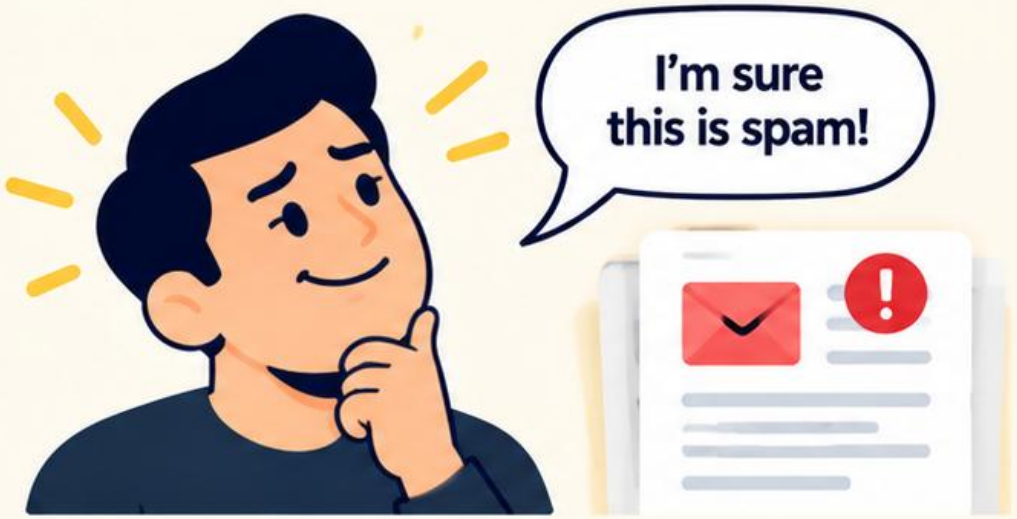
- Uses predefined options
- Shows probability scores
- Clear evaluation criteria
- Consistent and repeatable
- Easy to inspect and test

# Confidence Is Not Certainty

People can feel certain even when important information is missing — Jev AI provides probabilistic decisions and triggers human review when confidence is low.

## Human Judgment

Feels certain, but can be wrong




**Overconfidence**
High confidence even with limited information.

**WYSIATI**
What you see is all there is.
Missing context can lead to the wrong decision.

VS

## Jev AI

Probabilistic decision with human review




| | | |
|---|---|---|
| **High confidence** ≥ 0.80 | → | **Take action automatically** (e.g., mark as spam) |
| **Medium confidence** 0.50 – 0.79 | → | **Add more checks** (e.g., request more information) |
| **Low confidence** < 0.50 | → | **Send to human review** (e.g., analyst or user) |

# Why Losses Feel Bigger Than Gains

Loss aversion makes people react more strongly to losses than to equivalent gains.
Jev AI can classify risks objectively, while company policy decides how to handle costly mistakes.

## Human Psychology

Losses hurt more than gains of the same size

**Gain**

**+$100**

feels good



Positive feeling, but moderate impact

**Loss**

**−$100**

feels much worse



Stronger negative feeling, larger psychological impact

## Same Amount, Different Impact

Humans perceive losses as more significant than equivalent gains.




Feels **~2× stronger** than gains

Feels **less intense** than losses

## Jev AI in Practice

Objective risk classification + company policy

**Business case**

**−$100**

**Risk classification**

**High Risk**

Probability: **0.72**

**Company policy**

- Human review
- Additional checks
- Mitigation plan

## Key Takeaway

People tend to react more strongly to losses than to equal gains, even when the amounts are the same.
With Jev, we can use objective, probabilistic risk assessment, while company policy determines how to handle decisions with high potential cost.

# The Peak-End Rule — Why Customer Experience Can Be Misleading

People often remember an experience disproportionately through its emotional peak and ending.
Jev AI evaluates individual interactions, not just a final satisfaction rating.

## Human Memory

Many good moments, but the ending dominates






**Final memory:**

**"It was a bad experience."**

Even though most of the journey was positive.

## Jev AI Approach

Evaluate each interaction, not just the final rating

| | |
|---|---|
| Browse products | Positive |
| Add to cart | Positive |
| Checkout | Positive |
| Shipping | Positive |
| Product quality | Positive |
| Support | Positive |
| Delivery issue | Negative |



**Overall Analysis**

- 6 positive interactions
- 1 negative interaction
- Key issue: delivery
- Recommended action: Proactive resolution

- More accurate understanding
- Identifies the real issue
- Actionable insights
- Does not rely only on final rating

## Key Takeaway

A single negative ending can overshadow many positive moments in human memory.
With Jev, we analyze each interaction to get a complete and fair understanding of the customer experience.

# The Outside View — Why Base Rates Matter

People often focus on a compelling individual case and overlook statistical information from similar cases.
Jev AI helps evaluate decisions using real-world base rates and historical outcomes.

## Human Intuition

**A convincing individual story can be misleading**




### Individual Case

- Long-time customer
- Good payment history
- Explains a special situation
- Sounds reasonable and trustworthy

**May lead to approval**

Focuses on a single story, ignoring what usually happens in similar cases.

## The Outside View

Look at real-world data from similar cases

### Historical data (1,000 similar requests)



**85%** were approved

**15%** were rejected

### Outcome after approval

| | |
|---|---|
| No issues (likely safe) | 90% |
| Later problems (default/fraud) | 10% |

**Use base rates for a balanced view**

The outcome of similar cases provides valuable evidence for your decision.

## Jev AI Analysis

Combines case details with base rates

| | |
|---|---|
| Base rate (historical data) | 40% |
| Customer history | 25% |
| Current request details | 20% |
| Risk indicators | 15% |

**Risk classification** — Medium Risk

0.32 probability

**Recommended action**

**Send for human review**

Not high risk, but extra checks are advised given the base rate and current factors.

# From Fast Decisions to Responsible Automation

Fast judgments are useful, but some situations require deliberate checking.
Jev AI combines fast classification with business rules and human oversight.

## 1. Customer Request

Real-world input

I want a refund for my order. The package has not arrived.

- Order details
- Customer history
- Previous tickets
- Related events

## 2. Jev AI Classification

Fast, probabilistic decision



**Prediction: Valid Refund**

| | |
|---|---|
| Valid Refund | 0.72 |
| Possible Fraud | 0.20 |
| Insufficient Info | 0.08 |

**Key factors**

- Late delivery (12 days)
- Customer history: good
- Similar cases: high approval rate

## 3. Confidence Check and Business Rules

Decide the next step

**Confidence score**




**Business rules**

- Order older than 7 days
- No previous refunds
- Customer in good standing
- Item marked as delivered late

## 4. Action

Automatic or human review

**≥ 0.80** High confidence → **Take action automatically**
e.g., issue refund and notify customer

**0.50 – 0.79** Medium confidence → **Send to human review**
e.g., check additional details before decision

**< 0.50** Low confidence → **Investigate further**
e.g., possible fraud or missing information

## Key Takeaway

The best workflow does not automate every decision.
It uses fast AI predictions, clear business rules, and human oversight to make accurate and trustworthy decisions.



# Jev was named after who?

William Stanley Jevons FRS was an English economist and logician. Irving Fisher described Jevons's book The Theory of Political Economy as the start of the mathematical method in economics. It made the case that economics, as a science concerned with quantities, is necessarily mathematical.



# What is Jev?

A non-generative AI model that evaluates data and returns fast, probabilistic decisions instead of generating text.




Fast decisions

Probabilistic outputs

Type-safe and reliable

No text generation

# How Does Jev Work?

Jev takes your data, evaluates it using a non-generative model, and returns fast, probabilistic decisions in a structured format.

## 1 Input

Your data or query

| | |
|---|---|
| transaction_id | TXN_45821 |
| amount | $1,250 |
| merchant | Online Store |
| country | US |
| card_type | credit |
| ... | ... |

## 2 Jev

Evaluates your data

- Pattern analysis
- Learned signals
- Probabilistic modeling

Jev

TypeSafe AI

## 3 Probabilistic Decision

Structured, type-safe output

| | | |
|---|---|---|
| Fraud | | 72% |
| Legit | | 21% |
| Review | | 7% |

```
{
   "decision": "fraud",
   "probabilities": {
      "fraud": 0.72,
      "legit": 0.21,
      "review": 0.07
   }
}
```

### 1 Data

Structured or unstructured data is provided to Jev.

### 2 Probabilistic evaluation

Jev quickly evaluates the input using a non-generative model.

### 3 Decision

Jev returns fast, type-safe probabilistic results.

# Jev vs. LLMs

LLMs generate human-like text, while Jev evaluates data and returns fast, probabilistic decisions in a structured, type-safe format.

## LLMs

Generate human-like text

**Input**

Is this transaction suspicious?

Amount: $1,250
Merchant: Online Store
Country: US
Card type: credit




**Output**

This transaction could be suspicious because the amount is relatively high and the merchant is an online store. However, more context is needed to make a final decision.

**Generates free-form text**
Natural language responses

**Flexible and open-ended**
Can explain, write, summarize, etc.

**Not type-safe**
Output format can vary

## Jev

Evaluates data and returns probabilistic decisions

**Input**

Amount: $1,250
Merchant: Online Store
Country: US
Card type: credit




**Output**

```
{
  "decision": "fraud",
  "probabilities": {
    "fraud": 0.72,
    "legit": 0.21,
    "review": 0.07
  }
}
```

**Returns probabilistic decisions**
Structured and quantified results

**Type-safe output**
Predictable and consistent format

**Fast and optimized for decisions**
Built for evaluation, not text generation

# Jev

TypeSafe AI

AI-powered decisions with typed, probabilistic outputs.

VS

# n8n

Workflow Automation

Connect applications and execute automated steps.

| Jev | | n8n |
| --- | --- | --- |
| **AI-powered decisions**<br>Returns structured, probabilistic decisions for your data. | **Primary purpose** | **Workflow automation**<br>Automates processes by connecting multiple applications. |
| **Evaluate and classify inputs**<br>Analyzes information and returns typed results (e.g. classification, routing, risk scoring). | **Main function** | **Connect and execute steps**<br>Runs actions based on rules, triggers, and workflow logic. |
| **Core functionality**<br>Designed specifically for fast, reliable AI decisions (non-generative). | **AI capabilities** | **Uses integrated AI models and agents**<br>Supports LLMs and other AI tools as part of automated workflows. |

| Feature | Jev TypeSafe AI | n8n Workflow Automation |
| --- | --- | --- |
| Output | <pre>{<br>  "decision": "shipping",<br>  "probabilities":{<br>  "shipping": 0.92,<br>  "billing": 0.06<br>  }<br>}</pre> Typed decisions and probabilities | Workflow results and actions |
| Integrations |   Through APIs and application code | Extensive built-in integrations |
| Typical use | Classification, Routing, Risk scoring | Emails, Databases, CRM, APIs |

| Feature | Jev<br>TypeSafe AI | n8n<br>Workflow Automation |
| --- | --- | --- |
| Can use LLMs? | **No**<br>Designed as a non-generative alternative for decisions.<br>  | **Yes**<br>Supports LLM-based workflows.<br>  |
| Can automate tasks? | **No**<br>Not a complete automation platform.<br>  | **Yes**<br>Full workflow automation platform.<br>  |

# What Is System One AI?

System One AI makes fast, intuitive evaluations,
while deliberate reasoning takes more time and careful analysis.

## System One AI

Fast and intuitive evaluation



**Fast decisions**

Instant, probabilistic results

## Deliberate Reasoning

Slower and more analytical



**Slower decisions**

Step-by-step, detailed analysis

# Probabilistic Outputs

Jev returns probabilities for each possible outcome,
so you can understand the confidence and uncertainty.

## Input



**Transaction data**

e.g. amount, merchant,
country, card type, etc.

**Jev**

TypeSafe AI

## Probabilistic Output

| | | |
|---|---|---|
| Fraud | | 72% |
| Legit | | 21% |
| Review | | 7% |



**Confidence: 72%**

There is still uncertainty
across other possible outcomes.

# Type-Safe Decisions

Jev returns structured, predictable results with a defined format, so you always get the same type of output.

## Input



Transaction data

```
amount:      1250
merchant:    Online Store
country:     US
card_type:   credit
```








## Type-Safe Output


```
{
   "decision": "fraud",
   "probabilities": {
      "fraud": 0.72,
      "legit": 0.21,
      "review": 0.07
   },
   "confidence": 0.72,
   "reason_code": "HIGH_AMOUNT",
   "risk_level": "high"
}
```

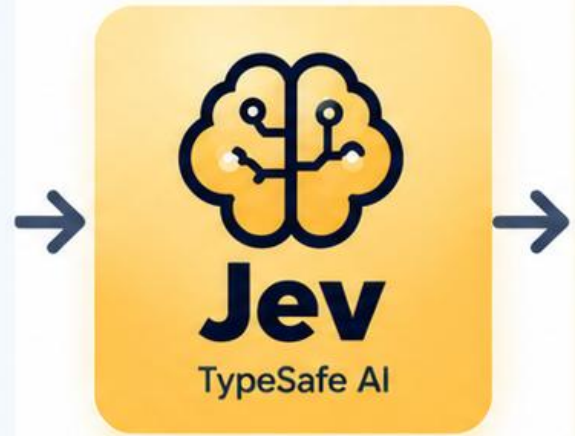
# Jev in Python

From input to typed output

## 1. Python code (input)

```python
from typesafe_sdk import Choice, TypeSafeClient

with TypeSafeClient() as client:
   result = client.system_one(
     state="My order has not arrived",
     questions={"route": Choice(
       instructions="Which team?",
       criteria={"shipping": "Delivery issues",
                 "billing": "Payment issues"}
     )})
   )
```




## 2. Typed output

```
result.choices["route"].choice
```

↓

**"shipping"**

Illustrative output

Install the SDK

```
pip install typesafe-sdk
```

# Jev in Action

## Building a Smart Support Router

### 1. Customer Message

A user sends a support request.

"My order hasn't arrived yet. Can you check where it is?"

Customer

### Jev

Evaluates the message using System One AI.

### 2. Routed to the Right Team

Jev predicts the most likely category and routes the request.

| | | |
|---|---|---|
| Selected | Shipping | 92% |
| | Billing | 6% |
| | Technical | 2% |

# Jev in Action

## Intelligent Email Prioritization

### Incoming Email

From: customer@example.com
Subject: **Server is down!**

Hi team,

**Our production server is down. Customers cannot access the website!**

Please help urgently.

Thank you.

### Jev

TypeSafe AI

Evaluates the email and predicts the priority with probabilities.

### Typed Output (Priority)

Selected

Urgent 94%

Normal 5%

Low 1%

# Jev in Action

## Detecting Suspicious Logins

### New Login Attempt

Correct password, but unusual behavior.

**New device**
Windows laptop

**Unfamiliar location**
Moscow, Russia

**Unusual time**
2:37 AM (your local time)

### Jev

Evaluates the login risk

Selected

Suspicious 96%

Normal 4%

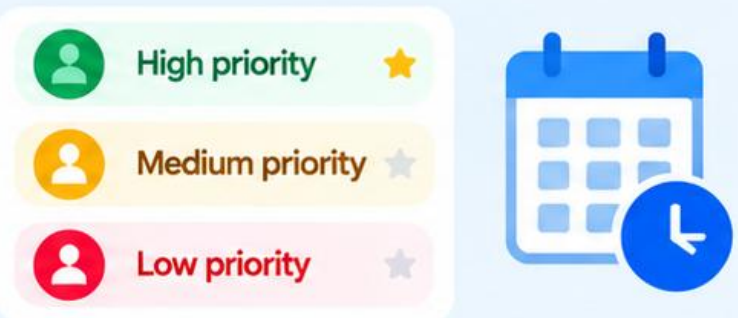


### Request Additional Verification

Instead of immediately granting account access.

Send Verification Code

# Practical Business Applications

| Business need | Jev's decision | Workflow action |
| --- | --- | --- |
|  **Customer support** Automatically route incoming customer messages. |  Jev   Billing, shipping, or technical? |   Assign the appropriate team. |
|  **Sales leads** Prioritize and focus on the most promising leads. |  Jev   High, medium, or low interest? |   Prioritize salesperson follow-up. |

# Practical Business Applications

| Business need | Jev's decision | Workflow action |
| --- | --- | --- |
| **Customer complaints**<br>Automatically identify urgent complaints. | Jev<br>Urgent 92%<br>Routine 8%<br>Urgent or routine? | **Escalate urgent complaints.**<br>Create a high-priority ticket for the support team. |
| **Invoice processing**<br>Automatically check incoming invoices. | Jev<br>Needs review 85%<br>No review needed 15%<br>Does this invoice need review? | **Send to accounting.**<br>Route invoices that need review to the accounting team. |
| **Customer feedback**<br>Automatically analyze customer feedback. | Jev<br>Positive 78%<br>Neutral 15%<br>Negative 7%<br>Positive, negative, or neutral? | **Update CRM and alert relevant staff.**<br>Save the feedback and notify the appropriate team. |

# How a Business Owner Can Get Started

## 1 Choose one repetitive decision

For example, categorizing incoming customer messages.

- I have a billing issue with my order. → Billing
- Where is my shipment? → Shipping
- The product is not working. → Technical

## 2 Connect the data source

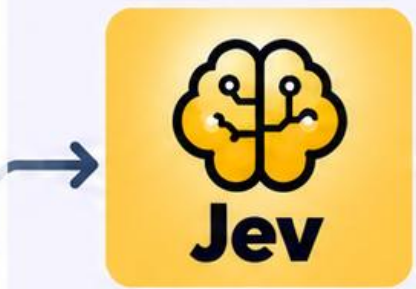
Use your existing systems such as a website, n8n, or a small Python service.

- Website
- Email
- CRM
- n8n
- Python

→ Your Application or n8n

## 3 Call the Jev API

Send the relevant information and a clearly defined question.




```
{
  "question": "Which category
   does this message belong to?",
  "choices": ["Billing", "Shipping",
   "Technical"],
  "input": "I have a billing issue
   with my order."
}
```

- Billing 92%
- Shipping 6%
- Technical 2%

# How a Business Owner Can Get Started

## 4 Apply business rules to the result.

Uncertain or sensitive decisions should go to human review.

| | | |
|---|---|---|
| High confidence 92% | → | Auto approve (Continue) |
| Uncertain 46% | → | Send to human review |
| Sensitive – | → | Require manual approval |

## 5 Automate the approved action.

Such as creating a ticket or assigning a salesperson.




- Create support ticket (Customer support)
- Assign salesperson (Sales team)
- Send notification (Email / Slack)
- Update CRM (Customer record)

## 6 Measure the results

Use real examples to evaluate accuracy and business impact before expanding automation.

### Example metrics

| 87% | 60% | 40% |
|---|---|---|
| Accuracy | Faster response time | Less manual work |

### Review real examples

- 20 correctly routed messages
- 3 escalated to human review (correctly)
- 1 misclassification (analyze and improve)